FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – X
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**B**	24.	**C**	47.	**B**
2.	**C**	25.	**C**	48.	**C**
3.	**C**	26.	**C**	49.	**A**
4.	**A, C**	27.	**A, B, D**	50.	**A, C, D**
5.	**A, C, D**	28.	**A, B, C**	51.	**B, C**
6.	**A, C, D**	29.	**A, B, C, D**	52.	**A, C, D**
7.	**A, C**	30.	**A, B, D**	53.	**A, D**
8.	**B, C**	31.	**B, D**	54.	**A, B, C**
9.	**A**	32.	**A**	55.	**D**
10.	**C**	33.	**B**	56.	**A**
11.	**5**	34.	**2**	57.	**4**
12.	**6**	35.	**4**	58.	**9**
13.	**6**	36.	**1**	59.	**4**
14.	**2**	37.	**3**	60.	**3**
15.	**4**	38.	**4**	61.	**5**
16.	**6**	39.	**2**	62.	**1**
17.	**8**	40.	**0**	63.	**1**
18.	**4**	41.	**5**	64.	**4**
19.	**4**	42.	**7**	65.	**6**
20.	**4**	43.	**2**	66.	**5**
21.	**00001.50**	44.	**45123.00**	67.	**00000.70**
22.	**00011.15**	45.	**31536.20**	68.	**00016.40**
23.	**00000.25**	46.	**00000.55**	69.	**00000.50**

Physics

PART – I

SECTION – A

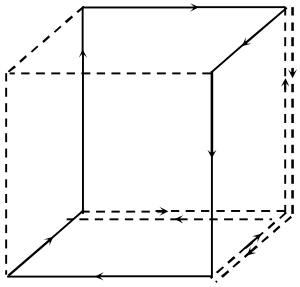
1. $g_{equator} = (0.9)g_{poles}$

$\Rightarrow \quad g_{pole} - \omega^2 R = (0.9)g_{pole}$

$\Rightarrow \quad \omega^2 R = \dfrac{g}{10}$

Peripheral velocity on equator $v = \omega R = \sqrt{\dfrac{g}{10R}} \times R = \sqrt{\dfrac{2gR}{20}} = \dfrac{v_e}{2\sqrt{5}}$

2. The current distribution is equivalent to three current carrying loops as shown in the figure.



3. When the rod moves towards left with velocity v and has displacement x towards left from mean position

$\varepsilon = B\ell v$

$q = C\varepsilon = CB\ell v$

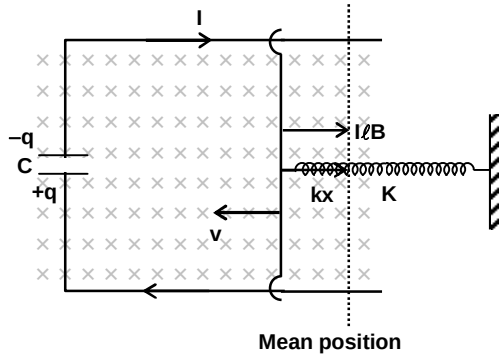
$I = \dfrac{dq}{dt} = CB\ell \dfrac{dv}{dt}$

$-(I\ell B + kx) = m\dfrac{dv}{dt}$

$\Rightarrow \quad -\left(CB^2\ell^2 \dfrac{dv}{dt} + kx\right) = m\dfrac{dv}{dt}$

$\Rightarrow \left(m + CB^2\ell^2\right)\dfrac{dv}{dt} = -kx$

$\Rightarrow \quad \omega = \sqrt{\dfrac{k}{m + CB^2\ell^2}} = \sqrt{\dfrac{2k}{3m}}$



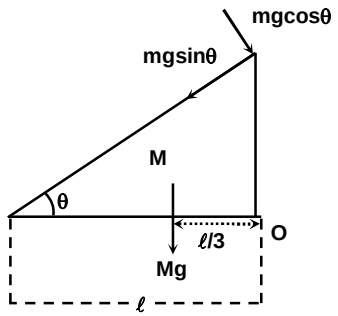
4. When $\mu > \tan\theta$, the particle will remain at rest on the wedge and force exerted by it on the wedge will be equal to mg in the vertically downward direction and wedge will remain at rest. When $\mu < \tan\theta$ the particle will slide down the wedge and force exerted by it on wedge will be as shown.

The wedge will topple if,

$\sum \tau_0 > 0$

$\Rightarrow \quad mg\cos\theta\,\ell\tan\theta\sin\theta - \mu mg\cos\theta\,\ell\tan\theta\cos\theta$

$-Mg\dfrac{\ell}{3} > 0$



$$\Rightarrow \quad m > \frac{M}{3\sin\theta\left(\sin\theta - \mu\cos\theta\right)}$$

5. From conservation of energy

$$-\frac{\sigma q}{2\varepsilon_0}\left(\sqrt{4R^2+x^2}-\sqrt{R^2+x^2}\right)=\frac{1}{2}mv^2-\frac{q\sigma R}{2\varepsilon_0}$$

$$\Rightarrow \quad v=\sqrt{\frac{q\sigma}{m\varepsilon_0}\left(R+\sqrt{R^2+x^2}-\sqrt{4R^2+x^2}\right)}$$

$$W_{\text{electric force}} = -\Delta U = -q\left(V_f - V_i\right)$$

$$=\frac{q\sigma}{2\varepsilon_0}\left(R+\sqrt{R^2+x^2}-\sqrt{4R^2+x^2}\right)$$

$$a=\frac{-qE}{m}=\frac{-q}{m}\frac{\sigma x}{2\varepsilon_0}\left(\frac{1}{\sqrt{R^2+x^2}}-\frac{1}{\sqrt{4R^2+x^2}}\right)$$

$$\simeq -\frac{q\sigma x}{4\varepsilon_0 mR} \text{ (since } x << R)$$

$$\Rightarrow \quad \omega=\sqrt{\frac{q\sigma}{4\varepsilon_0 mR}}$$

$$f=\frac{1}{2\pi}\sqrt{\frac{q\sigma}{4\varepsilon_0 mR}}$$

6. for circular orbit of satellite

$$E=\frac{U}{2}=-K \;\Rightarrow -dE=dK$$

$$\Rightarrow \quad (kv)vdt=mvdv$$

$$\frac{dv}{dt}=\frac{kv}{m} \;\Rightarrow\; \int dt=-\frac{m}{k}\int_{V_i}^{V_f}\frac{dv}{v}$$

$$\Rightarrow \quad t=\frac{m}{k}\ln\left(\frac{v_f}{v_i}\right)=\frac{m}{k}\ln\sqrt{\frac{GM/R}{GM/4R}}=\frac{m}{k}\ln 2$$

$$W_g = U_i - U_f = -\frac{GMm}{4R}-\left(-\frac{GmM}{R}\right)=\frac{3}{4}\frac{gR^2m}{R}=\frac{3}{4}mgR$$

$$W_g + W_r = \frac{1}{2}m\left(v_f^2 - v_1^2\right)=\frac{1}{2}m\left(\frac{GM}{R}-\frac{GM}{4R}\right)$$

$$\Rightarrow \quad \frac{3}{4}mgR + W_r = \frac{3}{8}mgR \Rightarrow\; W_r = \frac{-3}{8}mgR$$

7. The block will start moving
When $F > \mu mg$ i.e. $F > 10$ N
$\Rightarrow \quad t = 1$ s
The velocity of block is maximum when $F - \mu mg = 0$
i.e. at t = 5s
Net impulse = change in momentum

$$\Rightarrow \quad \frac{1}{2}6\times 30 - 2\frac{1}{2}10\times 1 - 10\times 4 = m v_{max}$$

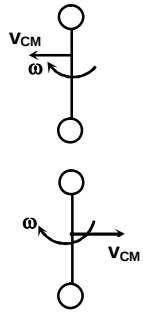
$$\Rightarrow \quad v_{max} = 20\,m/s$$

From $t = 1s$ to $t = 6s$

$$\frac{1}{2}\times 30\times 6 - 5 - 10\times 5 = 2v$$

$$\Rightarrow \quad v = \frac{35}{2} m/s$$

8. Let velocity of centre of mass of each dumbbell be V_{CM} and angular velocity ω as shown.



Since $\frac{v_2 - v_1}{u_1 - u_2} = 1 \Rightarrow \frac{\left(V_{CM} + \omega\frac{\ell}{2}\right) + \left(\frac{\omega\ell}{2} + V_{CM}\right)}{u - (-u)} = 1$

$$\Rightarrow \quad 2v_{CM} + \omega\ell = 2u \qquad \ldots(i)$$

Let the dumbbells exert average impulsive force N on each other during collision.

$$N\Delta t = 2m(V_{CM} + u)$$

$$N\Delta t\frac{\ell}{2} = \frac{m\ell^2}{2}\omega$$

$$\Rightarrow \quad \ell = \frac{\omega\ell^2}{2(V_{CM} + u)} \qquad \ldots(ii)$$

From equation (1) & (ii)

$$\omega = \frac{2u}{\ell}$$

9. $$\frac{m}{2}\sqrt{2g\ell} = mv_1 \Rightarrow v_1 = \sqrt{\frac{g\ell}{2}}$$

$mg\ell(1-\cos\theta) = \frac{1}{2}m\frac{g\ell}{2}$ (θ is angular displacement of the bob when it comes to rest)

$$\Rightarrow \cos\theta = \frac{3}{4}$$

When the bob moves in horizontal circle

$$\omega = \sqrt{\frac{g}{\ell\cos\theta}} = \sqrt{\frac{4g}{3\ell}}$$

$$v = \omega r = \omega\ell\sin\theta = \sqrt{\frac{4g}{3\ell}}\frac{\sqrt{7}}{4}\ell = \sqrt{\frac{7g\ell}{12}}$$

$$mv_Q = 2m\sqrt{\frac{7g\ell}{12}} \Rightarrow v_Q = \sqrt{\frac{7g\ell}{3}}$$

10. Height of the bob when string slacks

$$h = \ell + \ell(1-\cos\theta) = \ell + \frac{\ell}{4} = \frac{5\ell}{4}$$

Time $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{5\ell}{2g}}$

SECTION – C

11. $\mu\sin\theta = 1\times\sin 90°$

$\sin\theta = \frac{1}{\mu}$

$\Rightarrow \tan\theta = \frac{1}{\sqrt{\mu^2-1}} = \frac{1}{\sqrt{1+e^{x/d}-1}} = e^{-x/2d}$

$\frac{dy}{dx} = e^{-x/2d}$

$\Rightarrow y = 2d\left(1-e^{-x/2d}\right)$

At B, y = d

$\Rightarrow x = 2d\ell n2 = \frac{50}{7}\times 0.7 = 5m$

12. For the wedge $N\sin 30° = Ma$ …(i)

The acceleration of block perpendicular to the inclined plane is asin30°

$\Rightarrow mg\sin 60° - N = ma\sin 30°$ …(ii)

The net force on the block perpendicular to its direction of motion w.r.t ground should be zero.

$\Rightarrow N\cos 30° = mg\sin 30°$ …(iii)

$\Rightarrow$ N putting the value of N in equation (i) and (ii)

$\left(mg\tan 30° = Ma\right)$

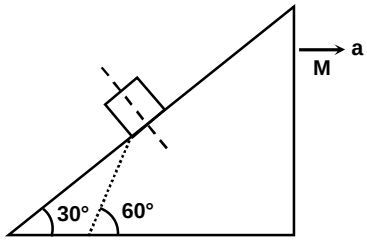
…(iv)

$mg\sin 60° - mg\tan 30° = ma\sin 30°$

…(v)

Equating the value of a in the equation (iv) and (v)

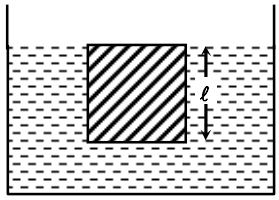
$\frac{m}{M} = 2 \Rightarrow m = 6kg$



13. $a = \frac{\rho_w A\left(\ell - x\right)g - \frac{\rho_w}{5}A\ell g}{\frac{\rho_w}{5}A\ell}$

$\int_0^v vdv = \int_0^\ell \left(4 - \frac{5x}{\ell}\right)gdx$

$\Rightarrow v = \sqrt{3\ell g} = \sqrt{3\times 1.2\times 10} = 6\ m/s$



14. For refraction at first surface of the lens,

$\frac{1.5}{-60} - \frac{4/3}{u} = \frac{1.5-\frac{4}{3}}{60}$

$\Rightarrow u = -48$ cm

If the object is placed h cm above the water surface, after refraction at the water surface its image will be formed at distance $\frac{4h}{3}$ from the surface

$\Rightarrow \frac{4h}{3} + 30 = 48 \Rightarrow h = \frac{27}{2}cm$

15. Induced current in resistor R

$$I = \frac{2Brv}{R}$$

Magnetic force on the ring $F_m = 2\frac{I}{2}(2r)B$

$$= \frac{4B^2r^2v}{R}$$

When the ring attains terminal velocity

$\sum F = 0$

$$\Rightarrow \quad mg - \frac{4B^2r^2v}{R} = 0$$

$$\Rightarrow \quad v = \frac{mgR}{4B^2r^2}$$

16. Since the loop is in equilibrium $\sum \tau = 0$

$$\Rightarrow \quad \left\{(2\lambda g\ell)\frac{\ell}{2}\sin\theta + \lambda\ell^2 g\sin\theta\right\}\hat{j} + I\ell^2(-\cos\theta\hat{i} - \sin\theta\hat{k}) \times B_0(\hat{i} - \hat{k}) = 0$$

$$\Rightarrow \quad B_0 = \frac{6\lambda g}{7I}$$

17. $\lambda_{k_\alpha} - \lambda_0' = 2\left(\lambda_{k_\alpha} - \lambda_0\right)$

$$\Rightarrow \quad \lambda_{k_\alpha} = 2\lambda_0 - \lambda_0'$$

$$\frac{\lambda_0'}{\lambda_0} = \frac{10}{20} = \frac{1}{2}$$

$$\Rightarrow \quad \lambda_{k_\alpha} = 2\lambda_0 - \frac{\lambda_0}{2} = \frac{3\lambda_0}{2}$$

$$\Rightarrow \quad \frac{4}{3R(z-1)^2} = \frac{3\lambda_0}{2}$$

$$\Rightarrow \quad z - 1 = \sqrt{\frac{8}{9\lambda_0 R}}$$

18. $$r = \frac{mv}{qB} = \frac{\sqrt{2mK_{max}}}{qB}$$

$$\Rightarrow K_{max} = \frac{q^2B^2r^2}{2m}$$

Putting the values of q, B, r and m

K_{max} = 2 eV

Energy of emitted photon

$h\nu = 13.6 \times 3^2\left(\frac{1}{9} - \frac{1}{16}\right)$eV = 5.95 eV

$h\nu = \phi + k_{max}$

$\Rightarrow \phi$ = 3.95 eV $\approx$ 4 eV

19. a = 4t

At t = 1 sec, a = 4 = μ_S g

$\Rightarrow \mu_S$ = 0.4

$\frac{dv}{dt} = 4t - \mu_k g$ (where v = velocity of block with respect to car)

$\Rightarrow \int_0^{v_2} dv = \left[2t^2 - \mu_k gt\right]_1^2$

$v_2 = 6 - \mu_k g$

From t = 2 s to t = 3s

v = u + at

$0 = (6 - \mu_K g) - \mu_K g \times 1$

$\Rightarrow \mu_K = 0.3$

$\Rightarrow \frac{3\mu_s}{\mu_k} = 4$

20. From conservation of mechanical energy

$-pE(\cos 90° - \cos 0°) = \frac{1}{2}\frac{mR^2}{2}\omega^2$

$(\lambda\pi R)\frac{4R}{\pi}E = \frac{1}{2}\frac{mR^2}{2}\omega^2$

$\omega = 4\left(\sqrt{\frac{\lambda E}{m}}\right)$

SECTION – D

21. $\nu = \nu_0 \frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2$ …(i)

$E = E_0 \frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4$

Solving (i) and (ii), n = 2 and z = 4

$L = mvr = \frac{nh}{2\pi}$, $\Delta L = \tau\Delta t = \Delta n\frac{h}{2\pi}$

$\tau = \frac{\Delta n}{\Delta t}\frac{h}{2\pi} = 1.5 \times 10^{-26}$ Nm

22. Zero error = $-(50 - 47) \times 0.01$ mm

$= -0.003$ cm

Measured diameter = $1.10 + 0.001 \times 12$

= 1.112 cm

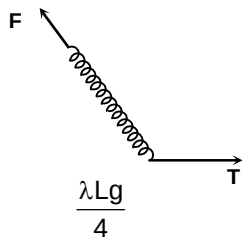
Actual diameter = Measured diameter – Zero Error

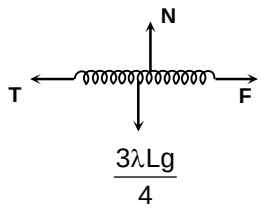
$= 1.112 - (-0.003) = 1.115$ cm = 11.15 mm

23. $F\cos 37° = \frac{\lambda L}{4}g$

$F\sin 37° = T = f \le \mu N \quad \left(N = \frac{3\lambda Lg}{4}\right)$

$\Rightarrow \mu \ge 1/4, \Rightarrow \mu_{min} = 1/4$



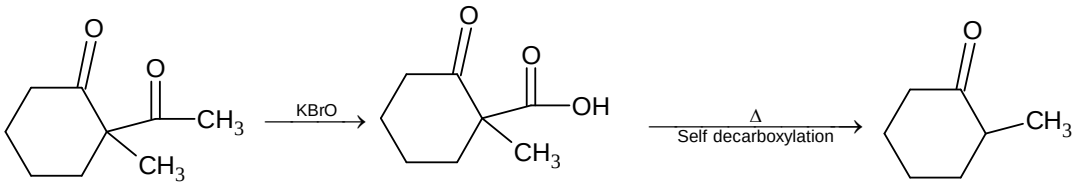


Chemistry

PART – II

SECTION – A

24. $8MnO_4^- + 3S_2O_3^{2-} + H_2O \longrightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

25. 

Haloform reaction

26. In this condition some atoms of H must be present in n = 3

So, ionization energy from n = 3, is $E = \frac{13.6}{n^2} = \frac{13.6}{9}$

= 1.51 V

27. Diastereomers have different physical properties, so, we can separate them by ordinary means such as recrystalization.

28. $NaOH(cold) + Cl_2 \longrightarrow NaCl + NaOCl + H_2O$

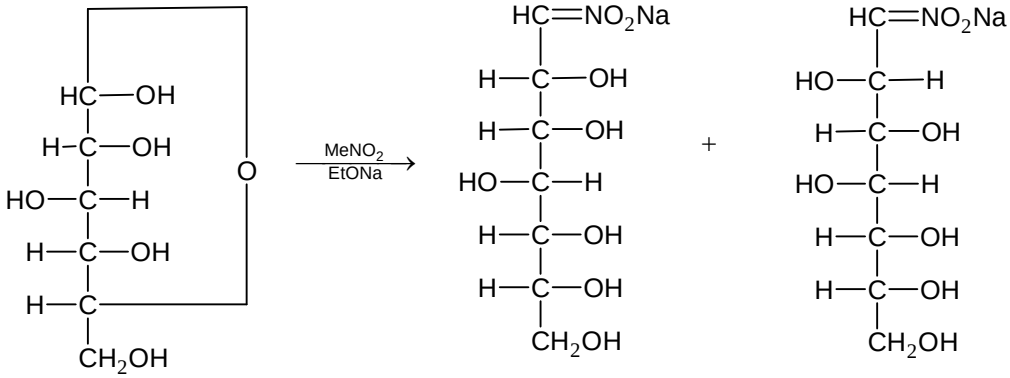
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Sodium hypochlorite

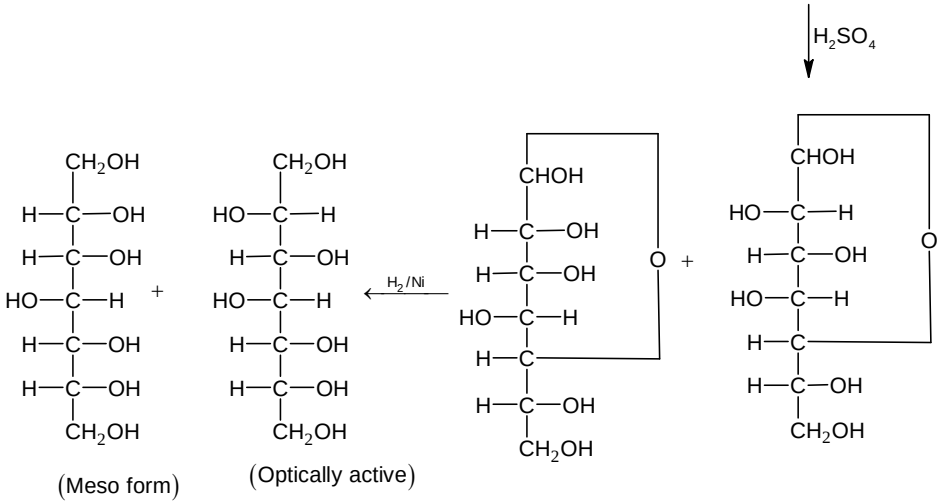
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anion $\overline{O}Cl$

contains σ bond

29. 

D – glucopyranose



30. If the temperature increases at constant pressure the gas expands, the mean free path increases, and the collision frequency decreases, so, A, B options are wrong.
For ideal gas, the average kinetic energy is dependent only on temperature, for isothermal change in pressure, the average kinetic energy would not change.

31. In electrophoresis and electroosmosis process motion of particles occur under applied charge.

32. $\frac{V}{T} = \text{constant (K)}$

$\log V = \log T + \log K$, where $K = \frac{nR}{P}$

For 2 graph

$\log \frac{nR}{P} = 0 = \log 1$

$\frac{nR}{P} = 1 \quad (R = P)$

n = 1 mol

for 1 graph

$\log \frac{nR}{P} = \log 3$

n = 3

log 3 graph

$\log \frac{nR}{P} = -\log 2 = \log \frac{1}{2}$

$n = \frac{1}{2}$ mol.

33. Straight line eqn

$(y - y_1) = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$

$$(P-16)=\left(\frac{4-16}{15-6}\right)(V-6)$$

$3P+4V=72$ For $T_{max}=\dfrac{(PV)_{max}}{nR}$

For $(PV)_{max}$

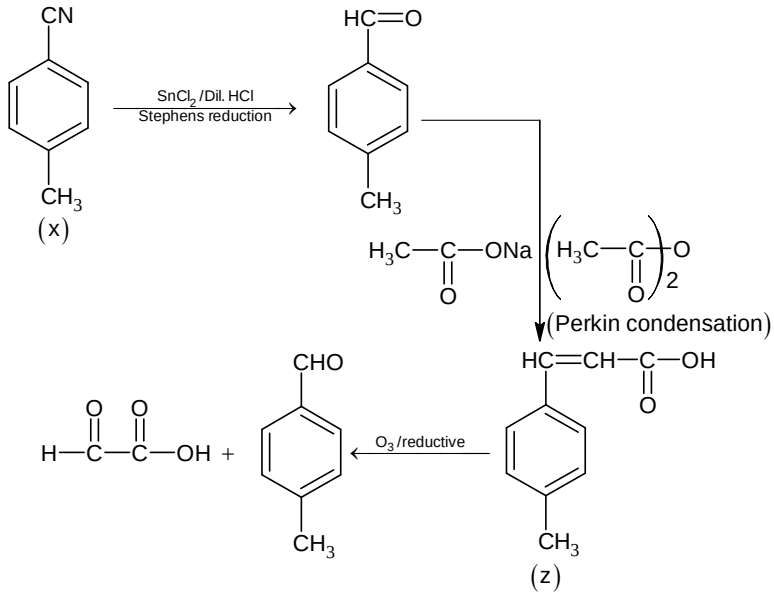
$3P=\dfrac{72}{2}$, $4V=\dfrac{72}{2}$

$P=12$ $V=9$

$\therefore T_{max}=\dfrac{12\times 9}{2\times\frac{1}{2}}=648$ K .

SECTION – C

34.



35. a, b, d because of unstable hydrate ‘c’ due to steric factor. ‘e’ gives a ketone.

36. $\left[M(AA)_3\right]\longrightarrow$ show optical isomerism

While in others plane of symmetry is possible.

37. $A(H_2O)_8(s)\rightleftharpoons \underset{(x+y)}{A(g)}+\underset{8x}{8H_2O(g)};$ K_{p_1}

$A(H_2O)_8(s)\rightleftharpoons \underset{(y+x)}{A(g)}+8H_2O(s);$ K_{p_2}

Given

$x+y=0.2$

$8x=0.001$

$K_{p_1}=(x+y)(8x)^8$

$=2\times10^{-25}$

$K_{p_2}=y+x$

$=0.2$

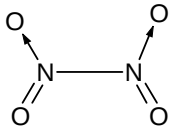
$H_2O(s) \rightleftharpoons H_2O(g)$

Vapour pressure $= \left(\frac{K_{p_1}}{K_{p_2}}\right)^{1/8}$

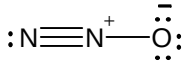
$= \left(\frac{2\times10^{-25}}{2\times10^{-1}}\right)^{1/8}$

$= 10^{-3}$ bar.

38. N_2O_4 is a dimer of NO_2.
It's correct structure is



The correct structure of N_2O is:



39. $r_{A_2} = K[A]^2$

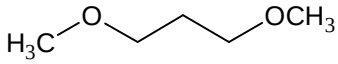
$K = \frac{r_{A_2}}{[A]^2} = \frac{10^{-5}}{(10^{-2})^2} = 10^{-1}$

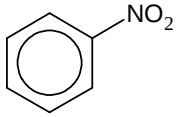
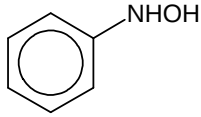
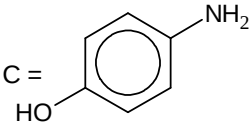
$\frac{K_A}{2} = K$

$K_A \Rightarrow 2K$

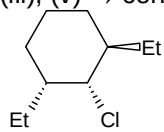
$= 2 \times 10^{-1}$

40. (i), (ii), (vi) $\rightarrow$ regiospecific reaction
(ii), (iii), (iv), (v), (vii) $\rightarrow$ stereospecific reaction.

41. Product is 

42. A =  ; B =  ; C =  ;

43. (ii) , (iv) $\rightarrow$ incorrect
(i), (iii), (v) $\rightarrow$ correct

(i)  is more reactive

By an E_2 mechanism due to presence of trans diaxial hydrogen. Anti periplannar transition state is possible.

(iii) In triethyl amine, pyramidal inversion is responsible for decreasing its nucleophilicity while in cyclic amine inversion is not possible.

SECTION – D

44. Sol. (1) $\underset{0.2\,M}{HCl} \longrightarrow H^+ + Cl^-$

Effective molarity = 0.4 M

Sol.(2) $BOH \longrightarrow B^+ + OH^-$

$t = 0$	0.2	0	0
$t = t$	$0.2 - x$	x	x

$K_b = \dfrac{x^2}{(0.2 - x)}$

$10^{-2} = \dfrac{x^2}{0.2 - x}$

x = 0.04

Effective molarity = 0.2 – x + x + x

= 0.2 + x

= 0.24

Sol.(3) $HCl + BOH \longrightarrow BCl + H_2O$

	0.2	0.2	0
moles	0	0	0.2

$\therefore$ Conc. of [BCl] $= \dfrac{0.2}{2} = 0.1\,M$

Effective molarity = 0.2 M

Sol. (4) effective molarity = 0.2 × 5 = 1.0

Sol. (5) effective molarity = 0.2 × 4 = 0.8

ΔT_f order $= 4 > 5 > 1 > 2 > 3$

And T_f order = 4 < 5 < 1 < 2 < 3

Hence the correct code is 45123.00

45. At anode:

$2Hg(\ell) + 2Cl^-(aq) \longrightarrow Hg_2Cl_2(s) + 2e^-$ … (1)

At cathode:

$HgO(s) + H_2O(\ell) + 2e^- \longrightarrow Hg(\ell) + 2OH^-(aq)$ … (2)

From Eq. (1) and (2)

Cell reaction

$HgO(s) + H_2O(\ell) + Hg(\ell) + 2Cl^-(aq)$

$\downarrow$

$Hg_2Cl_2(s) + 2OH^-(aq)$

$\Delta G = -nFE$

$= -2 \times 96500 \times (0.1634\ V)$

= 31536.2 J mol^{-1}.

46. % of unoccupied sites $= \dfrac{2.18 - 2.16801}{2.18} \times 100$

$= \dfrac{0.01199}{2.18} \times 100 = 0.55\%$

Mathematics

PART – III

SECTION – A

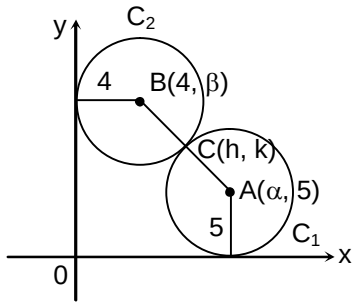
47. C(h, k) divides AB into the ratio 5 : 4

$\therefore (h,k)=\left(\frac{4\alpha+20}{9},\frac{5\beta+20}{9}\right)$

AB = 9 $\Rightarrow (\alpha-4)^2+(\beta-5)^2=9^2$

$\Rightarrow \frac{(h-4)^2}{4^2}+\frac{(k-5)^4}{5^2}=1$

Locus of point of contact is $\frac{(x-4)^2}{16}+\frac{(y-5)^2}{25}=1$



We shift the axes to (4, 5), so that curve become $\frac{x^2}{16}+\frac{y^2}{25}=1$ and point (9, 5) become (5, 0)

Chord of contact is $x=\frac{16}{5}$

Solving with conic, we get Q and R as $\left(\frac{16}{5},\pm 3\right)$

$\therefore$ Area of $\Delta PQR = \frac{27}{5}$

48. $\frac{AI}{A'I}=\frac{b+c}{a} \Rightarrow \frac{AI}{AA'}=\frac{b+c}{a+b+c}$

Similarly, $\frac{BI}{BB'}=\frac{a+c}{a+b+c}$ and $\frac{CI}{CI'}=\frac{a+b}{a+b+c}$

$\Rightarrow \frac{AI\cdot BI\cdot CI}{AA'\cdot BB'\cdot CC'}=\frac{(b+c)(a+c)(a+b)}{(a+b+c)^3}$

Now, $\frac{\frac{b+c}{a+b+c}+\frac{c+a}{a+b+c}+\frac{a+b}{a+b+c}}{3}\geq\left(\frac{(a+b)(b+c)(c+a)}{(a+b+c)^3}\right)^{1/3}$

$\Rightarrow \frac{(a+b)(b+c)(c+a)}{(a+b+c)^3}\leq\frac{8}{27}$

49. $a^2e^2 = 36 \Rightarrow a^2 - b^2 = 36$

Using $r = (s-a)\tan\frac{A}{2}$ in ΔOCF

1 = (s – a) (a = CF = OB)

$\Rightarrow$ 2 = 2s – 2a

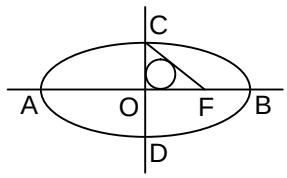
$\Rightarrow$ 2 = 2s – AB

$\Rightarrow 2=6+\frac{AB}{2}+\frac{CD}{2}-AB$

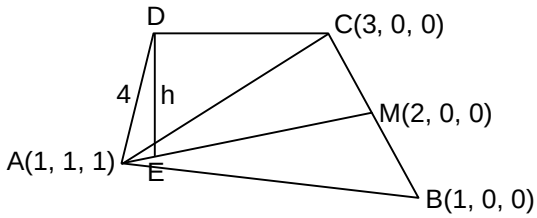
$\Rightarrow$ AB – CD = 8

$\Rightarrow$ a – b = 4 also $a^2 - b^2 = 36$

$\Rightarrow$ 2a = 13; 2b = 5



50. $v = \dfrac{2\sqrt{2}}{3}$



$$\Rightarrow \frac{1}{3}\cdot\frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 1 & 1\\ 2 & -1 & -1\end{vmatrix} h = \frac{2\sqrt{2}}{3}$$

$\Rightarrow h = 2$

Also ΔABC is right angled

Now let E divides AM in the ratio $\lambda : 1$

$E = \left(\dfrac{2\lambda+1}{\lambda+1}, \dfrac{1}{\lambda+1}, \dfrac{1}{\lambda+1}\right)$ and $AE^2 + DE^2 = AD^2$

$$\Rightarrow \left(\frac{2\lambda+1}{\lambda+1}-1\right)^2 + \left(1-\frac{1}{\lambda+1}\right)^2 + \left(1-\frac{1}{\lambda+1}\right)^2 + 4 = 16$$

$\Rightarrow$ Two values of λ

51. $\text{Re}\sum_{r=1}^{2016}\alpha^r = \sum_{r=1}^{2016}\cos\frac{r\pi}{2017} = 0$

$$\sum_{r=1}^{2016}\left(\alpha^r + \alpha^{-r}\right) = 2\sum_{r=1}^{2016}\cos\frac{r\pi}{2017} = 0$$

52. Equation of tangent to the hyperbola at P is $x\cdot\dfrac{1}{\alpha} + y\alpha = 2$

$\therefore$ $Q \equiv (2\alpha, 0)$ and $R \equiv \left(0, \dfrac{2}{\alpha}\right)$

Equation of normal to the hyperbola at P is $y - \dfrac{1}{\alpha} = \alpha^2(x-\alpha)$

$\therefore$ $M \equiv \left(\dfrac{\alpha^4-1}{\alpha^3}, 0\right)$ and $N \equiv \left(0, \dfrac{1-\alpha^4}{\alpha}\right)$

Now, area of ΔOQR = area of ΔOMN

$\Rightarrow (1-\alpha^4)^2 = 4\alpha^4$

$\Rightarrow \alpha^2 = \sqrt{2}-1$

53. Let roots are $\dfrac{a}{r^2}, \dfrac{a}{r}, a, ar, ar^2$

Now, $a\left(\dfrac{1}{r^2}+\dfrac{1}{r}+1+r+r^2\right) = 40$

$$\frac{1}{a}\left(r^2+r+1+\frac{1}{r}+\frac{1}{r^2}\right) = 10$$

Dividing both $\Rightarrow a^2 = 4 \Rightarrow a = \pm 2$

As product of roots $= a^5 = -\delta$

$\Rightarrow \delta = 32$ or -32

54. $\dfrac{\sin x dy - y\cos x dx}{\sin^2 x} = -\dfrac{1}{x^2}dx$

$$\Rightarrow d\left(\frac{y}{\sin x}\right) = -\frac{1}{x^2}dx$$

$$\Rightarrow y = \frac{\sin x}{x} + c\sin x$$

$\because\ y \to 0,\ x \to \infty \Rightarrow c = 0$

$f(x) = \dfrac{\sin x}{x}$

55. Let $z \in A$ and $\omega \in B$ then $z = e^{\frac{i2k_1\pi}{18}}$, $\omega = e^{\frac{i2k_2\pi}{48}}$ $k_1, k_2 \in I$

$\therefore\ z\omega = e^{\frac{i2\pi}{6}\left(\frac{k_1}{3}+\frac{k_2}{8}\right)} = e^{i2\left(\frac{8k_1+3k_2}{144}\right)\pi} = e^{\frac{i2k\pi}{144}}$ where $k = 8k_1 + 3k_2$

$\therefore\ z\omega$ can have 144 different values

56. Let $C_k = e^{\frac{i(2k-1)\pi}{n}}$, k = 1, 2, ..., n then $|C_k - C_1| = |\cos\theta + i\sin\theta - 1|$

where $\theta = \dfrac{(2k-1)\pi}{n} = 2\sin\dfrac{\theta}{2}$

$\therefore\ |C_2 - C_1| + |C_3 - C_1| + ... + |C_n - C_1| = 2\left[\sin\dfrac{\pi}{n} + \sin\dfrac{2\pi}{n} + ... + \sin\dfrac{(n-1)\pi}{n}\right]$

$= 2\cot\dfrac{\pi}{2n} = 2\cot\dfrac{180^o}{2\times 144} = 2\cot\left(\dfrac{5}{8}\right)^\circ$

SECTION – C

57. $\displaystyle\int_0^1\left(\frac{f(x)}{x}\right)^2 dx - 2\int_0^1 f(x)\,dx + \int_0^1 x^2 dx = 0$

$\Rightarrow \displaystyle\int_0^1\left(\frac{f(x)}{x} - x\right)^2 dx = 0$

$\Rightarrow f(x) = x^2$

58. If even letters goes to even envelopes but none goes to correct envelop, then the number of ways

$= 3!\left(1 - \dfrac{1}{1!} + \dfrac{1}{2!} - \dfrac{1}{3!}\right) = 2$

Now for $L_1\ L_2\ L_5\ L_7$

$E_1\ E_3\ E_5\ E_2$

Total number of without restriction = 4! = 24

Again if L_1, L_3, L_5 goes to their correct envelope number of ways = 1

If any two of them goes to their correct envelope, number of ways = ${}^3C_2 \times 1 = 3$

If any one goes to correct envelope, number of ways = ${}^3C_1 \times 3 = 9$

Total number of required ways for odd letters = 24 – 13 = 11

$\Rightarrow$ Required ways = $11 \times 2 = 22 = N$

59. $|z_2 + iz_1| = |z_1| + |z_2|$

$\Rightarrow z_1$ is perpendicular to z_2

Also, let $z_3 = z_1 + \dfrac{z_2 - z_1}{1-i}$

$\Rightarrow z_2 - z_3 = i(z_1 - z_3)$

$\Rightarrow \angle ACB = \dfrac{\pi}{2}$ and $|z_2 - z_3| = |z_1 - z_3|$

$\Rightarrow AC = BC$

$\Rightarrow AB^2 = AC^2 + BC^2$

$\Rightarrow AC^2 = \dfrac{AB^2}{2}$

$\Rightarrow$ Area of $\Delta ABC = \frac{1}{2}\cdot\frac{AB^2}{2} = \frac{7+9}{4} = 4$

60. $\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k}+\sqrt{k}} < \frac{2}{\sqrt{k}+\sqrt{k-1}} = 2\left(\sqrt{k}-\sqrt{k-1}\right)$

Putting k = 2, 3, ..., 10,000 and adding $\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+.....+\frac{1}{\sqrt{10,000}} < 2\left(\sqrt{10,000}-\sqrt{1}\right)$

$\Rightarrow A < 2\times 99 \Rightarrow A < 198$

Similarly $\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k}+\sqrt{k}} > \frac{2}{\sqrt{k+1}+\sqrt{k}} = 2\left(\sqrt{k+1}-\sqrt{k}\right)$

$\Rightarrow A > 2\left\{\sqrt{10,000}-\sqrt{2}\right\} > 197$

$\therefore [A] = 197$

$\therefore \left[\frac{[A]}{50}\right] = 3$

61. $2^m + 2^n + 2^p = (3-1)^m + (3-1)^n + (3-1)^p = 3\lambda + (-1)^m + (-1)^n + (-1)^p$, $\lambda \in I$

$\therefore$ Number of possible triplets = $50\times 25\times 12 + 50\times 25\times 13 = 31250$

62. Let $\cos^{-1}x = a$, $\sin^{-1}y = b$, $a \in [0, \pi]$, $b \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$a + b^2 = \frac{p\pi^2}{4}$, $a\cdot b^2 = \frac{\pi^4}{16}$, $b^2 \in \left[0, \frac{\pi^2}{4}\right]$

$\Rightarrow a + b^2 \in \left[0, \pi + \frac{\pi^2}{4}\right]$

$\Rightarrow 0 \le \frac{p\pi^2}{4} \le \pi + \frac{\pi^2}{4} \Rightarrow O \le p \le \frac{4}{\pi} + 1$

$\therefore$ p = 0, 1 or 2 but if p = 0, a = b = 0, not true

Also, $a\left(\frac{p\pi^2}{4} - a\right) = \frac{\pi^4}{16}$

$\Rightarrow 16a^2 - 4p\pi^2 a + \pi^4 = 0$

$\because a \in R$, $D \ge 0 \Rightarrow 16p^2\pi^4 - 64\pi^4 \ge 0 \Rightarrow p^2 \ge 4 \Rightarrow p \ge 2$

$\therefore$ Possible value of p is 2

63. $f(x)\cdot f(y(f(x)) = f(x + y)$

$\Rightarrow f'(x)\cdot f(yf(x)) + f(x)\cdot f'(yf(x))\cdot yf'(x) = f'(x + y)$

Putting x = 0 and then replacing y by x

$\Rightarrow -f(x) - xf'(x) = f'(x)$ ($\because$ f(0) = 1, f'(0) = –1)

$\Rightarrow \frac{f'(x)}{f(x)} = -\frac{1}{1+x}$

Integrating $\ln|f(x)| = -\ln|1 + x| + \ln|c|$

$\Rightarrow f(x) = \frac{c}{1+x} \Rightarrow f(x) = \frac{1}{1+x}$ ($\because$ f(0) = 1)

$\therefore$ Required area = $\int_0^\infty \frac{1}{(1+x)^2}dx = 1$

64. Let $e^x = t$

$$\therefore I = \int_1^2 \frac{2t^3+t^2-1}{t^3+t^2-t+1}\frac{dt}{t} = \int_1^2 \frac{t\left(3t^2+2t-1\right)-\left(t^3+t^2-t+1\right)}{t\left(t^3+t^2-t+1\right)}dt$$

$$= \int_1^2 \frac{3t^2+2t-1}{t^3+t^2-t+1} - \int_1^2 \frac{dt}{t} = \left[\ln\left|t^3+t^2-t+1\right|\right]_1^2 - \left[\ln|t|\right]_1^2 = \ln\frac{11}{4}$$

65. Eccentricity of the hyperbola in ${}^nC_r$
Possible value of ${}^nC_r$ will be ${}^2C_1$, ${}^3C_1$, ${}^4C_1$, ${}^5C_1$, ${}^4C_2$ and ${}^5C_2$

66. $AB = I$
$(AB)C_1 = C_1$, $(AB)^2C_2 = C_2$ ..... and so on
$t_r(C_r) = r\cdot 3^r + (r-1)\cdot 3^r = (2r-1)3^r$

$$\sum_{r=1}^{50} t_r\left((AB)^r\ C_r\right) = t_r\left((AB)C_1\right) + t_r\left((AB)^2 C_2\right) + ... + t_r\left((AB)^{50}\ C_{50}\right)$$

$= t_r(C_1) + t_r(C_2) + ... + t_r(C_{50})$
$= 1\cdot 3' + 3\cdot 3^2 + 5\cdot 3^3 + ... + 99\cdot 3^{50}$ (AGP)
$\Rightarrow S = 3 + 49\cdot 3^{51}$
$\therefore a + b = 100$

SECTION – D

67. Let roots be $\alpha_1, \alpha_2, ..., \alpha_8$

$\alpha_1 + \alpha_2 + ... + \alpha_8 = 4$, $\alpha_1\alpha_2 ..... \alpha_8 = \frac{1}{2^8}$

$\therefore$ A.M. and G.M. of $\alpha_1, \alpha_2, ..., \alpha_8$ are equal

$\Rightarrow \alpha_1 = \alpha_2 = ... = \alpha_8 = \frac{1}{2}$

$\Rightarrow a_2 = {}^8C_6\cdot\left(\frac{1}{2}\right)^6$, $a_4 = {}^8C_4\cdot\frac{1}{2^4}$, $a_6 = {}^8C_2\cdot\frac{1}{2^2}$

68. Reflection of (4, 5) w.r.t lines y = x and y = 2x are (5, 4) and $\left(\frac{8}{5}, \frac{31}{5}\right)$ respectively minimum value of PA + PB + AB = distance between (5, 4) and $\left(\frac{8}{5}, \frac{31}{5}\right) = \sqrt{16.4}$

69. Let $g(x) = \sin^{-1}|\sin x| + \cos^{-1}(\cos x) = \begin{cases} 2x & ; \quad 0 \le x \le \frac{\pi}{2} \\ \pi & ; \quad \frac{\pi}{2} < x \le \frac{3\pi}{2} \\ 4\pi - 2x & ; \quad \frac{3\pi}{2} < x \le 2\pi \end{cases}$

g(x) is periodic with period 2π and is constant in the interval $\left[2n\pi+\frac{\pi}{2}, 2n\pi+\frac{3\pi}{2}\right]$, $n \in I$

Now $f(x) = g(\alpha x)$
$\therefore$ f(x) is constant in $\left[\frac{2n\pi}{\alpha}+\frac{\pi}{2\alpha}, \frac{2n\pi}{\alpha}+\frac{3\pi}{2\alpha}\right]$

$\therefore 2\pi = \frac{3\pi}{2\alpha} - \frac{\pi}{2\alpha}$

$\Rightarrow \alpha = \frac{1}{2}$